

Sensory Wardrobe — Entity-Relationship Diagram (FINAL)

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Narrative

This ERD represents the final logical data model for the Sensory Wardrobe application. It maps directly to the five SQLite data stores (DS1-DS5) identified in the Level 0 DFD.

Based on instructor feedback from the draft submission, this version introduces a proper junction table (OUTFIT_LOG_ITEM) to normalize the many-to-many relationship between outfit logs and wardrobe items, replacing the previous JSON array approach.

The central feedback loop — where users log outfits, rate comfort, and receive improved suggestions — is reflected in the relationships between WARDROBE_ITEM, OUTFIT_LOG, COMFORT_RATING, and WEATHER_SNAPSHOT.

ER Diagram

```
erDiagram
    USER_PROFILE {
```

```
    int id PK
    string display_name
    string email
    string password_hash
    boolean is_dependent
    int caregiver_id FK
    string sensory_preferences
    string role
    boolean is_disabled
    datetime created_at
    datetime updated_at
}
```

```
WARDROBE_ITEM {
    int id PK
    int user_id FK
    string name
    string category
    string fabric
    string sensory_tags
    int warmth_level
    string photo_path
    boolean is_active
    datetime created_at
    datetime updated_at
}
```

```
OUTFIT_LOG {
    int id PK
    int user_id FK
    date logged_date
    int weather_snapshot_id FK
    string notes
    datetime created_at
}
```

```
OUTFIT_LOG_ITEM {
    int id PK
    int outfit_log_id FK
    int wardrobe_item_id FK
}
```

```

}

COMFORT_RATING {
    int id PK
    int outfit_log_id FK
    int user_id FK
    int overall_score
    int texture_score
    int pressure_score
    int temperature_score
    datetime created_at
}

```

```

WEATHER_SNAPSHOT {
    int id PK
    float location_lat
    float location_lon
    float temperature_c
    float feels_like_c
    int humidity
    string condition
    float wind_speed
    datetime fetched_at
}

```

```

USER_PROFILE ||--o{ USER_PROFILE : "manages"
USER_PROFILE ||--o{ WARDROBE_ITEM : "owns"
USER_PROFILE ||--o{ OUTFIT_LOG : "logs"
USER_PROFILE ||--o{ COMFORT_RATING : "rates"
OUTFIT_LOG ||--|| COMFORT_RATING : "rated by"
OUTFIT_LOG ||--o{ OUTFIT_LOG_ITEM : "contains"
WARDROBE_ITEM ||--o{ OUTFIT_LOG_ITEM : "included in"
WEATHER_SNAPSHOT ||--o{ OUTFIT_LOG : "context for"

```

Note: The self-referencing relationship (USER_PROFILE manages USER_PROFILE) may not render in all Mermaid versions. See Relationship Summary below for documentation.

Entity Descriptions

Entity	Store	Purpose
USER_PROFILE	DS1	User accounts: caregivers, dependents, admins. Self-referencing caregiver_id enables multi-profile.
WARDROBE_ITEM	DS2	Clothing catalog with sensory tags and warmth level (1-5) for suggestion engine.
OUTFIT_LOG	DS3	Daily outfit selection linked to user, weather, and items via junction table.
OUTFIT_LOG_ITEM	DS3	Junction table resolving M:N between logs and items.
COMFORT_RATING	DS4	Post-wear comfort scores (overall required, sub-scores optional). Feeds suggestion ranking.
WEATHER_SNAPSHOT	DS5	Cached weather from OpenWeatherMap. Provides context for outfit logs and suggestions.

Relationship Summary

Relationship	Card.	FK Location
USER_PROFILE to USER_PROFILE	1:M	caregiver_id in child
USER_PROFILE to WARDROBE_ITEM	1:M	user_id in WARDROBE_ITEM

USER_PROFILE to OUTFIT_LOG	1:M	user_id in OUTFIT_LOG
USER_PROFILE to COMFORT_RATING	1:M	user_id in COMFORT_RATING
OUTFIT_LOG to COMFORT_RATING	1:1	outfit_log_id in COMFORT_RATING
OUTFIT_LOG to OUTFIT_LOG_ITEM	1:M	outfit_log_id in junction
WARDROBE_ITEM to OUTFIT_LOG_ITEM	1:M	wardrobe_item_id in junction
WEATHER_SNAPSHOT to OUTFIT_LOG	1:M	weather_snapshot_id in OUTFIT_LOG

Attribute Details

USER_PROFILE (DS1)

Attribute	Type	Constraints
id	INTEGER	PK, AUTO_INCREMENT
display_name	TEXT	NOT NULL
email	TEXT	NOT NULL, UNIQUE
password_hash	TEXT	NOT NULL
is_dependent	BOOLEAN	NOT NULL, DEFAULT FALSE
caregiver_id	INTEGER	FK, NULLABLE
sensory_preferences	TEXT	NULLABLE (JSON)
role	TEXT	NOT NULL, DEFAULT 'user'
is_disabled	BOOLEAN	NOT NULL, DEFAULT FALSE

created_at	DATETIME	NOT NULL
updated_at	DATETIME	NOT NULL

WARDROBE_ITEM (DS2)

Attribute	Type	Constraints
id	INTEGER	PK, AUTO_INCREMENT
user_id	INTEGER	FK, NOT NULL
name	TEXT	NOT NULL
category	TEXT	NOT NULL
fabric	TEXT	NULLABLE
sensory_tags	TEXT	NULLABLE (JSON array)
warmth_level	INTEGER	NOT NULL, CHECK 1-5
photo_path	TEXT	NULLABLE
is_active	BOOLEAN	NOT NULL, DEFAULT TRUE
created_at	DATETIME	NOT NULL
updated_at	DATETIME	NOT NULL

OUTFIT_LOG (DS3)

Attribute	Type	Constraints
id	INTEGER	PK, AUTO_INCREMENT
user_id	INTEGER	FK, NOT NULL
logged_date	DATE	NOT NULL
weather_snapshot_id	INTEGER	FK, NULLABLE
notes	TEXT	NULLABLE

created_at	DATETIME	NOT NULL
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OUTFIT_LOG_ITEM (Junction)

Attribute	Type	Constraints
id	INTEGER	PK, AUTO_INCREMENT
outfit_log_id	INTEGER	FK, NOT NULL
wardrobe_item_id	INTEGER	FK, NOT NULL

Composite UNIQUE on (outfit_log_id, wardrobe_item_id).

COMFORT_RATING (DS4)

Attribute	Type	Constraints
id	INTEGER	PK, AUTO_INCREMENT
outfit_log_id	INTEGER	FK, NOT NULL, UNIQUE
user_id	INTEGER	FK, NOT NULL
overall_score	INTEGER	NOT NULL, CHECK 1-5
texture_score	INTEGER	NULLABLE, CHECK 1-5
pressure_score	INTEGER	NULLABLE, CHECK 1-5
temperature_score	INTEGER	NULLABLE, CHECK 1-5
created_at	DATETIME	NOT NULL

WEATHER_SNAPSHOT (DS5)

Attribute	Type	Constraints
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id	INTEGER	PK, AUTO_INCREMENT
location_lat	REAL	NOT NULL
location_lon	REAL	NOT NULL
temperature_c	REAL	NOT NULL
feels_like_c	REAL	NOT NULL
humidity	INTEGER	NOT NULL
condition	TEXT	NOT NULL
wind_speed	REAL	NULLABLE
fetches_at	DATETIME	NOT NULL

Design Decisions

1. **Normalized M:N Relationship:** OUTFIT_LOG_ITEM junction table replaces the item_ids JSON array. Provides referential integrity, enables JOINS for per-item analytics, and supports CASCADE deletes.
2. **Self-Referencing FK:** caregiver_id points back to USER_PROFILE.id. When is_dependent = TRUE, the record belongs to a dependent managed by the referenced caregiver.
3. **1:1 Comfort Rating:** Enforced via UNIQUE constraint on outfit_log_id. Each outfit is rated once after wearing.
4. **Weather Snapshot Reuse:** Snapshots cached independently. Multiple logs from same day/location share one snapshot, reducing storage and API calls.
5. **Soft Delete:** is_active (items) and is_disabled (users) preserve history. Inactive items excluded from suggestions but visible in past logs.

6. **Tags as JSON String:** Pragmatic choice for SQLite + Flutter. Predefined tag list enforced at application layer.
 7. **Role-Based Access:** Three levels: user, caregiver, admin. Authorization enforced at router and repository layers.
-

Schema SQL

```
CREATE TABLE user_profiles (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    display_name TEXT NOT NULL,  
    email TEXT NOT NULL UNIQUE,  
    password_hash TEXT NOT NULL,  
    is_dependent INTEGER NOT NULL DEFAULT 0,  
    caregiver_id INTEGER  
        REFERENCES user_profiles(id),  
    sensory_preferences TEXT,  
    role TEXT NOT NULL DEFAULT 'user',  
    is_disabled INTEGER NOT NULL DEFAULT 0,  
    created_at TEXT NOT NULL,  
    updated_at TEXT NOT NULL  
);  
  
CREATE TABLE wardrobe_items (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    user_id INTEGER NOT NULL  
        REFERENCES user_profiles(id),  
    name TEXT NOT NULL,  
    category TEXT NOT NULL,  
    fabric TEXT,  
    sensory_tags TEXT,  
    warmth_level INTEGER NOT NULL  
        CHECK(warmth_level BETWEEN 1 AND 5),  
    photo_path TEXT,  
    is_active INTEGER NOT NULL DEFAULT 1,  
    created_at TEXT NOT NULL,
```

```

        updated_at TEXT NOT NULL
    );

CREATE TABLE weather_snapshots (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    location_lat REAL NOT NULL,
    location_lon REAL NOT NULL,
    temperature_c REAL NOT NULL,
    feels_like_c REAL NOT NULL,
    humidity INTEGER NOT NULL,
    condition TEXT NOT NULL,
    wind_speed REAL,
    fetched_at TEXT NOT NULL
);

CREATE TABLE outfit_logs (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    user_id INTEGER NOT NULL
        REFERENCES user_profiles(id),
    logged_date TEXT NOT NULL,
    weather_snapshot_id INTEGER
        REFERENCES weather_snapshots(id),
    notes TEXT,
    created_at TEXT NOT NULL
);

CREATE TABLE outfit_log_items (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    outfit_log_id INTEGER NOT NULL
        REFERENCES outfit_logs(id)
        ON DELETE CASCADE,
    wardrobe_item_id INTEGER NOT NULL
        REFERENCES wardrobe_items(id),
    UNIQUE(outfit_log_id, wardrobe_item_id)
);

CREATE TABLE comfort_ratings (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    outfit_log_id INTEGER NOT NULL UNIQUE
        REFERENCES outfit_logs(id)

```

```
        ON DELETE CASCADE,  
user_id INTEGER NOT NULL  
        REFERENCES user_profiles(id),  
overall_score INTEGER NOT NULL  
        CHECK(overall_score BETWEEN 1 AND 5),  
texture_score INTEGER  
        CHECK(texture_score BETWEEN 1 AND 5),  
pressure_score INTEGER  
        CHECK(pressure_score BETWEEN 1 AND 5),  
temperature_score INTEGER  
        CHECK(temperature_score BETWEEN 1 AND 5),  
created_at TEXT NOT NULL  
);
```

FINAL — Normalization improvements based on instructor draft feedback. Submitted as part of Deliverable 3.